NEET Botany Golden Set v1.0

90 curated mixed-style questions for RAG problem-solving evaluation

Instructions: Each question has four options. Answers and solutions are intentionally not included in this question-only version.

Q1.

Read the following statements and choose the set
of correct statements.
In the members of Phaeophyceae.
A. Asexual reproduction occurs usually by
biflagellate zoospores.
B. Sexual reproduction is by oogamous method
only.
C. Stored food is in the form of carbohydrates
which is either mannitol or laminarin
D. The major pigments found are chlorophyll a,
c and carotenoids and xanthophyll.
E.
Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.
Choose the correct answer from the options given
below:
(1) A, C, D and E only
(2) A, B, C and E only
(3) A, B, C and D only
(4) B, C, D and E only

Q2.

Bulliform cells are responsible for
(1) Increased photosynthesis in monocots.
(2) Providing large spaces for storage of sugars.
(3) Inward curling of leaves in monocots.
(4) Protecting the plant from salt stress.

Q3.

How many different proteins does the ribosome consist of?
a. 20
b. 80
c. 60
d. 40

Q4.

Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production?
a. Abscisic Acid
b. Indole-3-butyric Acid
c. Gibberellic Acid
d. Zeatin

Q5.

Given below are two statement: One is labelled as Assertion (A) and the other is labelled as
Reason (R).

Assertion (A): Both wind and water pollinated
flowers are not very colourful and do not produce nectar.
Reason (R): The flowers produce enormous
amount of pollen grains in wind and water pollinated flowers.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both A and R are true and R is the correct
explanation of A.
(2) Both A and R are true but R is NOT the correct
explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

Q6.

Match List I with List II List I List II
A.
Two or more alternative forms of a gene I. Back cross
B.
Cross of F1 progeny with homozygous recessive parent II. Ploidy
C.
Cross of F1 progeny with any of the parents III. Allele
D.
Number of chromosome sets in plant IV. Test cross
Choose the correct answer from the options given
below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-III, D-IV

Q7.

Tropical regions show greatest level of species richness because
A. Tropical latitudes have remained relatively
undisturbed for millions of years, hence more
time was available for species diversification.
B. Tropical environments are more seasonal.
C. More solar energy is available in tropics.
D. Constant environments promote niche
specialization.
E. Tropical environments are constant and
predictable.
Choose the correct answer from the options given
below:
(1) A, B and E only
(2) A, B and D only
(3) A, C, D and E only
(4) A and B only

Q8.

Which micronutrient is required for splitting of water molecule during photosynthesis?

a. copper

b. manganese

c. molybdenum

d. magnesium

Q9.

A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?

(1) Only pink flowered plants

(2) Red, Pink as well as white flowered plants

(3) Only red flowered plants

(4) Red flowered as well as pink flowered plants

Q10.

The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?

a. Diakinesis

b. Pachytene

c. Zygotene

d. Diplotenes

Q11.

The correct sequence of events in the life cycle of bryophytes is

A. Fusion of antherozoid with egg.

B. Attachment of gametophyte to substratum.

C. Reduction division to produce haploid spores.

D. Formation of sporophyte.

E.

Release of antherozoids into water.

Choose the correct answer from the option given

below:

(1) D, E, A, C, B

(2) B, E, A, C, D

(3) B, E, A, D, C

(4) D, E, A, B, C

Q12.

Family Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens, pick out the characteristics specific to family Fabaceae but not found in Solanaceae or Liliaceae.

a. Epiphyllous and Dithecous anthers

b. Diadelphous and Dithecous anthers

c. Polyadelphous and epipetalous stamens

d. Monoadelphous and Monothecous anthers

Q13.

The DNA present in chloroplast is:

(1) Linear, single stranded

(2) Circular, single stranded

(3) Linear, double stranded

(4) Circular, double stranded

Q14.

Which of the following are required for the dark reaction of photosynthesis?

A. Light

B. Chlorophyll

C. CO_2

D. ATP

E. NADPH

Choose the correct answer from the options given

below:

(1) C, D and E only

(2) D and E only

(3) A, B and C only

(4) B, C and D only

Q15.

Identify the incorrect statement related to Pollination:

A. Moths and butterflies are the most dominant pollinating agents among insects

B. Pollination by water is quite rare in flowering plants

C. Pollination by wind is more common amongst abiotic pollination

D. Flowers produce foul odours to attract flies and beetles to get pollinated

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Q16.

Unequivocal proof that DNA is the genetic material was first proposed by

a. Wilkins and Franklin

b. Fedrick, Griffith

c. Alfred Hershey and Martha Chase

d. Avery, Macleoid and McCarthy

Q17.

Which of the following answers are NOT correct?

a. The amount of some toxic substances of industrial waste

water increases in the organisms at successive trophic levels.

b. The micro-organisms involved in biodegradation of

organic matter in a sewage polluted water body consume a lot of oxygen causing the death of aquatic organisms.

c. Algal blooms caused by excess of organic matter in water

improve water quality and promote fisheries.

d. Water hyacinth grows abundantly in eutrophic water

bodies and leads to an imbalance in the ecosystem dynamics of the water body.

Q18.

Match List-I with List-II

List-I

List-II

(A)

Iron (I) Synthesis of auxin

(B)

Zinc (II) Component of nitrate reductase

(C)

Boron (III) Activator of catalase

(D)

Molybdenum (IV) Cell elongation and differentiation

Choose the correct answer from the options given below:

a. A- II, B- IV, C- I, D-III b. A-III, B- II, C-I, D-IV

c. A-III, B- III, C-IV, D-I d. A-III, B-I, C-IV, D-II

Q19.

Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence

A. Insertion of recombinant DNA into the host cell.

B. Cutting of DNA at specific location by restriction enzyme.

C. Isolation of desired DNA fragment.

D. Amplification of gene of interest using PCR.

Choose the correct answer from the option given below:

a. B, D, A, C

b. B, C, D, A

c. C, A, B, D

d. C, B, D, A

Q20.

Cellulose does not form blue colour with Iodine because

a. It breaks down when iodine reacts with it.

b. It is a disaccharide.

c. It is a helical molecule.

d. It does not contain complex helices and hence cannot hold

iodine molecules.

Q21.

Match List I with List II List I List II

A.

Rhizopus I. Mushroom

B.

Ustilago II. Smut fungus

C.

Puccinia III. Bread mould

D.
Agaricus IV. Rust fungus
Choose the correct answer from the options given
below:
(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-I, B-III, C-II, D-IV

Q22.

Given below are two statements: One is labelled as Assertion A and the other is labeled as Reason R:
Assertion A: A flower is defined as modified shoot wherein
the shoot apical meristem changes to floral meristem.
Reason R: Internode of the shoot gets condensed to produce
different floral appendages laterally at successive nodes instead of leaves.
In the light of the above statements, choose the correct answer from the options given below:
a. A is false but R is true.
b. Both A and R are true and R is the correct explanation of A.
c. Both A and R are true but R is NOT the correct explanation
of A.
d. A is true but R is false.

Q23.

Lecithin, a small molecular weight organic compound found in living tissues, is an example of:
(1) Glycerides
(2) Carbohydrates
(3) Amino acids
(4) Phospholipids

Q24.

Given below are two statements :
Statement I: The forces generated by transpiration can lift a
xylem-sized column of water over 130 meters height.
Statement II: Transpiration cools leaf surfaces sometimes
10 to 15 degrees, by evaporative cooling.
In the light of the above statements, choose the most appropriate answer from the options given below:
a. Statement I is incorrect but Statement II is correct.
b. Both Statement I and Statement II are correct.
c. Both Statement I and Statement II are incorrect.
d. Statement I is correct but Statement II is incorrect.

Q25.

Given below are two statements: One is labelled as
Assertion (A) and the other is labelled as Reason
(R).
Assertion (A): A typical unfertilized, angiosperm

embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R): The egg apparatus has 2 polar nuclei.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct

explanation of A.

(2) Both A and R are true but R is NOT the correct

explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

Q26.

Upon exposure to UV radiation, DNA stained with ethidium bromide will show

a. Bright orange colour

b. Bright red colour

c. Bright blue colour

d. Bright yellow colour

Q27.

Match List-I with List-II

List-I (Interaction)

List-II (Species A And B)

(A)

Mutualism (I) +(A), O(B)

(B)

Commensalism (II) -(A), O(B)

(C)

Amensalism (III) +(A), -(B)

(D)

Parasitism (IV) +(A), +(B)

Choose the correct answer from the given options:

a. A-III, B-I, C-IV, D-II

b. A-IV, B-II, C-I, D-III

c. A-IV, B-I, C-II, D-III

d. A-IV, B-III, C-I, D-II

Q28.

Movement and accumulation of ions across a membrane against their concentration gradient can be explained by

a. Active Transport

b. Osmosis

c. Facilitated Diffusion

d. Passive Transport

Q29.

During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out

a. Polysaccharides
b. RNA
c. DNA
d. Histones

Q30.

Match List-I with List-II.
List-I
List-II
(A) Nucleolus
(I) Site of formation of glycolipid
(B) Centriole
(II) Organization like the cartwheel
(C) Leucoplasts
(III) Site for active ribosomal RNA synthesis
(D) Golgi apparatus (IV)
For storing nutrients
Choose the correct answer from the options given
below
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-III, C-I, D-IV

Q31.

Gemmae are present in:
A. Pteridophytes
B. Some Gymnosperms
C. Some Liverworts
D. Mosses

Q32.

Given below are two statements:
Statement I: Endarch and exarch are the terms often used for
describing the position of secondary xylem in the plant body.
Statement II: Exarch condition is the most common feature
of the root system.
In the light of the above statements, choose the correct answer from the options given below:
a. Statement I is incorrect but Statement II is true.
b. Both Statement I and Statement II are true.
c. Both Statement I and Statement II are false.
d. Statement I is correct but Statement II is false.

Q33.

Among eukaryotes, replication of DNA takes place in
a. G2 phase

b. M phase

c. S phase

d. G1 phase

Q34.

How many molecules of ATP and NADPH are required for every molecule of CO_2 fixed in the Calvin cycle?

(1) 3 molecules of ATP and 3 molecules of

NADPH

(2) 3 molecules of ATP and 2 molecules of

NADPH

(3) 2 molecules of ATP and 3 molecules of

NADPH

(4) 2 molecules of ATP and 2 molecules of

NADPH

Q35.

What is the function of tassels in the corn cob?

a. To protect seeds

b. To attract insects

c. To trap pollen grains

d. To disperse pollen grains

Q36.

The lactose present in the growth medium of bacteria is transported to the cell by the action of

(1) Permease

(2) Polymerase

(3) Beta-galactosidase

(4) Acetylase

Q37.

The thickness of ozone in a column of air in the atmosphere is measured in terms of:

a. Kilobase

b. Dobson units

c. Decameter

d. Decibels

Q38.

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: ATP is used at two steps in glycolysis.

Reason R: First ATP is used in converting glucose into

glucose-6-phosphate and second ATP is used in conversion of fructose-6- phosphate into fructose-1-6-diphosphate.

In the light of the above statements, choose the correct answer from the options given below is true,

a. A is false but R is true.

b. Both A and R are true and R is the correct explanation

of A

c. Both A and R are true but R is NOT the correct explanation
of A.
d. A is true but R is false.

Q39.

In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.
a. Silver
b. Copper
c. Zinc
d. Tungsten or gold

Q40.

Spindle fibers attach to kinetochores of chromosomes during
(1) Anaphase
(2) Telophase
(3) Prophase
(4) Metaphase

Q41.

Each of the following characteristics represent a Kingdom proposed by Whittaker. Arrange the following in increasing order of complexity of body organization.
A. Multicellular heterotrophs with cell wall made
of chitin.
B. Heterotrophs with tissue/organ/organ system
level of body organization.
C. Prokaryotes
with cell wall made of
polysaccharides and amino acids.
D. Eukaryotic autotrophs with tissue/organ level
of body organization.
E.
Eukaryotes with cellular body organization.
Choose the correct answer from the options given
below:
(1) A, C, E, B, D
(2) C, E, A, D, B
(3) A, C, E, D, B
(4) C, E, A, B, D

Q42.

Identify the correct statements:
A. Lenticels are the lens-shaped openings permitting the
exchange of gases.
B. Bark formed early in the season is called hard bark.
C. Bark is a technical term that refers to all tissues exterior
to vascular cambium.

D. Bark refers to periderm and secondary phloem.
E. Phellogen is single-layered in thickness. Choose the
correct answer from the options given below:
a. B and C only
b. B. C and E only
c. A and D only
d. A, B and D only

Q43.

Given below are two Statements:
Statement I : Chromosomes become gradually visible
under light microscope during leptotene stage.
Statement II: The beginning of diplotene stage is
recognized by dissolution of synaptonemal complex.
In the light of the above statements, choose the correct answer from the options given below:
(1) Statement I is true but Statement II is false.
(2) Statement I is false but Statement II is true.
(3) Both Statement I and Statement II are true.
(4) Both Statement I and Statement II are false

Q44.

Which hormone promotes internode/petiole elongation in deep water rice?
a. 2, 4-D
b. GA3
c. Kinetin
d. Ethylene

Q45.

Which of the following is an example of actinomorphic flower?
(1) Pisum
(2) Sesbania
(3) Datura
(4) Cassia

Q46.

Hind II always cuts DNA molecules at a particular point called recognition sequence and it consists of:
(1) 4 bp
(2) 10 bp
(3) 8 bp
(4) 6 bp

Q47.

The equation of Verhulst-Pearl logistic growth is
dN K N rN . dt K − □ □ = □ □ □ □
From this equation, K indicates:
(1) Carrying capacity

(2) Population density
(3) Intrinsic rate of natural increase
(4) Biotic potential

Q48.

How many ATP and NADPH, are required for the synthesis of one molecule of Glucose during Calvin cycle?
a. 18 ATP and 16 NADPH
b. 12 ATP and 12 NADPH
c. 18 ATP and 12 NADPH
d. 12 ATP and 16 NADPH

Q49.

What is the fate of piece of DNA carrying only gene of interest which is transferred into an alien organism?
A. The piece of DNA would be able to multiply
itself independently in the progeny cells of the
organisms.
B. It may get integrated into the genome of the
recipient.
C. It may multiply and be inherited along with the
host DNA.
D. The alien piece of DNA is not an integrated
part of chromosome.
E.
It shows ability to replicate.
Choose the correct answer from the options given
below:
(1) B and C only
(2) A and E only
(3) A and B only
(4) D and E only

Q50.

Which of the following stages of meiosis involves division of centromere?
a. Telophase
b. Metaphase I
c. Metaphase II
d. Anaphase II

Q51.

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In gymnosperms the pollen grains are released
from the microsporangium and carried by air currents.
Reason R: Air currents carry the pollen grains to the mouth
of the archegonia where the male gametes are discharged and pollen tube is not formed.
In the light of the above statements, choose the correct answer from the options given below:
a. A is false but R is true.

b. Both A and R are true and R is the correct explanation of A.

c. Both A and R are true but R is NOT the correct explanation

of A.

d. A is true but R is false.

Q52.

Match List-I with List-II.

List-I

List-II

(A) Rose

(I) Twisted aestivation

(B) Pea

(II) Perigynous flower

(C) Cotton

(III) Drupe

(D) Mango

(IV) Marginal placentation

Choose the correct answer from the options given

below

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-III, C-IV, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-I, B-II, C-III, D-IV

Q53.

Match List-I with List-II

List-I

List-II

(A)

M Phase (I) Proteins are synthesized

(B)

G2 Phase (II) Inactive phase

(C)

Quiescent stage (III) Interval between mitosis and initiation of DNA replication

(D)

G1 Phase (IV) Equational division

Choose the correct answer from the options given below:

a. A-II, B-IV, C-I, D-III

b. A-III, B-II, C-IV, D-I

c. A-IV, B-II, C-I, D-III

d. A-IV, B-I, C-II, D-III

Q54.

Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield?

(1) Cytokinin

(2) Abscisic acid

(3) Auxin

(4) Gibberellin

Q55.

In some members of which of the following pairs of families, pollen grains retain their viability for months after release?

A. Poaceae; Leguminosae

B. Poaceae; Solanaceae

C. Rosaceae; Leguminosae

D. Poaceae; Rosaceae

Q56.

Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by

a. Henking

b. Thomas Hunt Morgan

c. Sutton and Boveri

d. Alfred Sturtevant

Q57.

Identify the correct statements:

A. Detrivores perform fragmentation

B. The humus is further degraded by some microbes during mineralization.

C. Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.

D. The detritus food chain begins with living organisms.

E. Earthworms break down detritus into smaller particles by a process called catabolism.

Choose the correct answer from the options given below:

a. D, E, A only

b. A, B, C only

c. B, C, D only

d. C, D, E only

Q58.

Match List-I with List-II

List-I

List-II

(A)

Cohesion (I) More attraction in liquid phase

(B)

Adhesion (II) Mutual attraction among water molecules

(C)

Surface tension (III) Water loss in liquid phase

(D)

Guttation (IV) Attraction towards polar surfaces

Choose the correct answer from the option given below:

a. A-II, B-I, C-IV, D-III

b. A-II, B-IV, C-I, D-III

c. A-IV, B-III, C-II, D-I

d. A-III, B-I, C-IV, D-II

Q59.

In a plant, black seed color (BB/Bb) is dominant over white seed color (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?

(1) Bb

(2) BB/Bb

(3) BB

(4) bb

Q60.

The cofactor of the enzyme carboxypeptidase is:

(1) Flavin

(2) Haem

(3) Zinc

(4) Niacin

Q61.

Given below are the stages in the life cycle of pteridophytes. Arrange the following stages in the correct sequence.

A. Prothallus stage

B. Meiosis in spore mother cells

C. Fertilisation

D. Formation of archegonia and antheridia in

gametophyte.

E. Transfer of antherozoids to the archegonia in

presence of water.

Choose the correct answer from the options given

below:

(1) B, A, D, E, C

(2) B, A, E, C, D

(3) D, E, C, A, B

(4) E, D, C, B, A

Q62.

Axile placentation is observed in

a. China rose, Petunia and Lemon

b. Mustard, Cucumber and Primorose

c. China rose, Beans and Lupin

d. Tomato, Dianthus and Pea

Q63.

Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of:

(1) Competitive inhibition

(2) Enzyme activation

(3) Cofactor inhibition

(4) Feedback inhibition

Q64.

Auxin is used by gardeners to prepare weed free lawns. But no damage is caused to grass as auxin;

(1) does not affect mature monocotyledonous plants.

(2) can help in cell division in grasses, to produce growth.

(3) promotes apical dominance.

(4) promotes abscission of mature leaves only.

Q65.

Identify the set of correct statements:

A. The flowers of Vallisneria are colourful and produce nectar

B. The flowers of waterlily are not pollinated by water.

C. In most of water-pollinated species, the pollen grains are protected from wetting.

D. Pollen grains of some hydrophytes are long and ribbon like.

E. In some hydrophytes, the pollen grains are carried passively inside water.

Choose the correct answer from the options given below:

(1) A, C, D and E only

(2) B, C, D and E only

(3) C, D and E only

(4) A B, C and D only

Q66.

What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

a. Transcription of only snRNAS

b. Transcription of rRNAs (28S, 18S and 5.8S)

c. Transcription of tRNA, 5 srRNA and snRNA

d. Transcription of precursor of mRNA

Q67.

These are regarded as major causes of biodiversity loss:

A. Over exploitation

B. Co-extinction

C. Mutation

D. Habitat loss and fragmentation

E.

Migration

Choose the correct option:

(1) A, B and E only

(2) A, B and D only

(3) A, C and D only

(4) A, B, C and D only

Q68.

Which of the following combinations is required for chemiosmosis?

a. proton pump, electron gradient, NADP synthase

b. membrane, proton pump, proton gradient. ATP synthase

c. membrane, proton pump, proton gradient, NADP synthase

d. proton pump, electron gradient, ATP synthase

Q69.

A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end;

(1) Inducer, Repressor, Structural gene

(2) Promoter, Structural gene, Terminator

(3) Repressor, Operator gene, Structural gene

(4) Structural gene, Transposons, Operator gene

Q70.

Malonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

a. Dinitrogenase

b. succinic dehydrogenase

c. Amylase

d. Lipase

Q71.

Genera like Selaginella and Salvinia produce two kinds of spores. Such plants are known as:

A. Heterosorus

B. Homosporous

C. Heterosporous

D. Homosorus

Q72.

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Late wood has fewer xylary elements with

narrow vessels.

Reason R: Cambium is less active in winters.

In the light of the above statements, choose the correct answer from the options given below:

a. A is false but R is true.
b. Both A and R are true and R is the correct explanation of A.
c. Both A and R are true but R is NOT the correct explanation
of A.
d. A is true but R is false.

Q73.

Match List-I with List-II
List-I
List-II
(A)
Oxidative decarboxylation (I) Citrate synthase
(B)
Glycolysis (II) Pyruvate
(C)
Oxidative phosphorylation (III) Electron transport system
(D)
Tricarboxylic acid cycle (IV) EMP pathway
Choose the correct answer from the options given below:
a. A-II, B-IV, C-III,, D-I
b. A-III, B-IV, C-II,, D-I
c. A-II, B-IV, C-I,, D-III
d. A-III, B-I, C-II,, D-IV

Q74.

In angiosperm, the haploid, diploid and triploid structures of a fertilized embryo sac sequentially are:
a. Synergids, antipodals and Polar nuclei
b. Synergids, Primary endosperm nucleus and zygote
c. Antipodals, synergids, and primary endosperm nucleus
d. Synergids, Zygote and Primary endosperm nucleus

Q75.

Which one of the following can be explained on the basis of Mendel's Law of Dominance?
A. Out of one pair of factors one is dominant and
the other is recessive.
B. Alleles do not show any expression and both
the characters appear as such in F2 generation.
C. Factors occur in pair in normal diploid plants.
D. The discrete unit controlling a particular
character is called factor.
E.
The expression of only one of the parental
characters is found in a monohybrid cross.
Choose the correct answer from the options given
below:
(1) B, C and D only (2) A, B, C, D and E

(3) A, B and C only (4) A, C, D and E only

Q76.

The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called;

(1) Semi-conservative method

(2) Sustainable development

(3) in-situ conservation

(4) Biodiversity conservation

Q77.

In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as:

a. Senescence

b. Differentiation

c. Dedifferentiation

d. Development

Q78.

Which of the following statements are correct about Klinefelter’s Syndrome?

A. This disorder was first described by Langdon Down (1866).

B. Such an individual has overall masculine development.

However, the feminine development is also expressed.

C. The affected individual is short stature.

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile.

Choose the correct answer from the options given below:

a. A and E only

b. A and B only

c. C and D only

d. B and E only

Q79.

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The first stage of gametophyte in the life cycle

of moss is protonema stage.

Reason R: Protonema develops directly from spores produced

in capsule.

In the light of the above statements, choose the most appropriate answer from the options given below:

a. A is not correct but R is correct.

b. Both A and R are correct and R is the correct explanation

of A.

c. Both A and R are correct but R is NOT the correct

explanation of A.

d. A is correct but R is not correct.

Q80.

Given below are two statements:

Statement I: Cleistogamous flowers are invariably autogamous.

Statement II: Cleistogamy is disadvantageous as there is no chance for cross pollination.

In the light of the above statements, choose the correct answer from the options given below.

A. Statement I is incorrect but Statement II is correct

B. Both Statement I and Statement II are correct

C. Both Statement I and Statement II are incorrect

D. Statement I is correct but Statement II is incorrect

Q81.

The phenomenon of pleiotropism refers to

a. more than two genes affecting a single character.

b. presence of several alleles of a single gene controlling a

single crossover.

c. presence of two alleles, each of the two genes controlling

a single trait.

d. a single gene affecting multiple phenotypic expression.

Q82.

In the equation GPP - R = NPP

GPP is Gross Primary Productivity

NPP is Net Primary Productivity.

R here is

a. Reproductive allocation

b. Photosynthetically active radiation

c. Respiratory quotient

d. Respiratory loss

Q83.

Expressed Sequence Tags (ESTs) refers to

a. Certain important expressed genes.

b. All genes that are expressed as RNA.

c. All genes that are expressed as proteins.

d. All genes whether expressed or unexpressed.

Q84.

Identify the pair of heterosporous pteridophytes among the following:

a. Equisetum and Salvinia b. Lycopodium and Selaginella

c. Selaginella and Salvinia d. Psilolum and Salvinia

Q85.

Large, colourful, fragrant flowers with nectar are seen in:

a. wind pollinated plants

b. insect pollinated plants

c. bird pollinated plants

d. bat pollinated plants

Q86.

Given below are two statements:

Statement I: Gause's 'Competitive Exclusive Principle'

states that two closely related species competing for the same related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II: In general, carnivores are more adversely

affected by competition than herbivores.

In the light of the above statements, choose the correct answer from the options given below:

a. Statement I is incorrect but Statement II is true

b. Both Statement I and Statement II are true

c. Both Statement I and Statement II are false

d. Statement I is correct but Statement II is false.

Q87.

Which one of the following is not a criterion for classification of fungi?

(1) Mode of spore formation

(2) Fruiting body

(3) Morphology of mycelium

(4) Mode of nutrition

Q88.

Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species?

a. Co-extinctions

b. Habitat loss and fragmentation

c. Over exploitation for economic gain

d. Alien species invasions

Q89.

Which of the following algae produce Carrageen?

A. Brown algae

B. Red algae

C. Blue-green algae

D. Green algae

Q90.

The historic Convention on Biological Diversity, 'The Earth Summit' was held in Rio de Janeiro in the year:

a. 2002

b. 1992

c. 1985

d. 1986

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